UNIT – 2
AGILE MINI-PROJECT

Student Record and Academic Management Platform

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1. Study of Business Problem and Product Vision

The project is a Student Record and Academic Management Platform. The purpose of the platform is to provide a web-based system for managing student records, attendance, academic performance, course registration, authentication, notifications and reports. The Unit 2 activities begin by reviewing the Product Discovery documents prepared in Unit 1.

Unit 1 information to be reviewed:

- Product Vision Statement
- Mission Statement
- Stakeholders
- Business Requirements
- Functional Requirements
- Non-functional Requirements
- User Personas
- Minimum Viable Product (MVP)
- Product Roadmap

2. Agile Product Vision

Product Objective: Develop a secure and user-friendly Student Record and Academic Management Platform that enables educational institutions to efficiently manage student records, attendance, academic performance, course registration and reporting through a scalable web-based application.

Item	Description
Primary Users	Students, Faculty Members, Administrative Staff and other authorized users.
Business Benefits	Centralized academic information, easier record management, accurate attendance and performance tracking, and efficient reporting.
Success Criteria	Core modules work correctly, authorized users can access relevant functions, records can be managed accurately, and reports can be generated.
Project Constraints	Limited development time, team capacity, implementation complexity and dependency between modules.
Expected Capabilities	Authentication, student management, course management, attendance, marks, notifications and reporting.

3. Scrum Team Structure

The project is organized as a Scrum Team with the following responsibilities.

Scrum Role	Responsibilities
Product Owner	Defines product requirements, prioritizes the Product Backlog, represents stakeholder interests and accepts completed work.
Scrum Master	Facilitates Scrum ceremonies, removes project impediments, ensures Agile practices are followed and supports team collaboration.

Development Team	Designs the application, writes code, tests software, prepares documentation and demonstrates completed features.

4. Project Epics

Epic	Features / Scope
Epic 1 – Student Management	Student Registration; Student Profile Management; Student Search; Student Update; Student Deletion
Epic 2 – Course Management	Add Course; Update Course; Assign Students; View Course Details
Epic 3 – Attendance Management	Record Attendance; View Attendance; Attendance Reports
Epic 4 – Marks and Academic Performance	Enter Marks; Update Marks; Calculate Grades; Generate Result Sheets
Epic 5 – Authentication and Security	User Login; Logout; Password Reset; Role-Based Access Control
Epic 6 – Notifications	Attendance Alerts; Examination Notifications; Result Notifications
Epic 7 – Reporting	Student Reports; Attendance Reports; Performance Reports; Course Reports

5. User Stories

ID	User Story	Priority	Story Points
US01	As a Student, I want to register and maintain my profile so that my academic information is available in the system.	High	5
US02	As an authorized user, I want to search student records so that I can quickly find required information.	High	3
US03	As a Faculty Member, I want to record student attendance so that attendance records remain accurate and updated.	High	5
US04	As a Faculty Member, I want to enter and update marks so that student academic performance is maintained.	High	5
US05	As a Faculty Member, I want to manage courses and assign students so that course information is organized.	Medium	5
US06	As a User, I want to log	High	5

	in securely so that only authorized users can access the system.		
US07	As a User, I want to receive examination, attendance and result notifications so that I stay informed.	Medium	3
US08	As an Administrator, I want to generate student and academic reports so that information can be reviewed efficiently.	Medium	5
US09	As a Faculty Member, I want to view attendance reports so that I can monitor attendance.	Medium	3
US10	As an Administrator, I want role-based access control so that users can access only appropriate functions.	High	8

6. Acceptance Criteria

Student Login

- Login page loads successfully.
- Registered users can log in.
- Invalid credentials display appropriate error messages.
- Successful login redirects to the dashboard.
- Session management functions correctly.

Record Attendance

- Faculty can select a course.
- Faculty can select a lecture date.
- Faculty can mark attendance.
- Attendance records are saved successfully.
- Attendance reports can be generated.

Add Student

- Authorized user can open the student registration form.
- Required student details can be entered.
- Valid student data can be saved.
- Saved student information can be viewed later.
- Invalid or incomplete input is handled appropriately.

Enter Marks

- Faculty can select a student/course.
- Marks can be entered.

- Marks can be updated.
- The saved marks are displayed correctly.
- Grades/results can be calculated or generated where applicable.

Generate Report

- Authorized user can select the required report.
- Relevant records are included.
- The report is generated successfully.
- The report is readable and usable.

7. Prioritized Product Backlog

ID	Backlog Item	Epic	Priority	Points
PB01	User Login	Authentication	High	5
PB02	Add Student	Student Management	High	5
PB03	Search Student	Student Management	High	3
PB04	Record Attendance	Attendance Management	High	5
PB05	Enter/Update Marks	Academic Performance	High	5
PB06	Add Course	Course Management	Medium	3
PB07	Attendance Report	Reporting	Medium	3
PB08	Attendance Alert	Notifications	Medium	3
PB09	Academic Analytics Dashboard	Reporting	Low	8

Backlog prioritization is based on business value, user importance, technical dependencies and implementation complexity.

8. Story Point Estimation

Story Points	Complexity
1	Very Easy
2	Easy
3	Moderate
5	Difficult
8	Very Difficult
13	Highly Complex

The team can use Planning Poker to collaboratively estimate the effort required for User Stories.

9. Sprint Planning

Sprint	Sprint Goal	Activities
Sprint 1	Develop user authentication and project setup.	Repository initialization; User registration; Login functionality; Basic user interface; Testing.
Sprint 2	Develop Student Management Module.	Add Student; Search Student; Update Student; Delete Student.

Sprint 3	Develop Course and Attendance Management.	Add Course; Assign Courses; Record Attendance; Attendance Reports.

10. Sprint Backlog

Sprint	Task	Owner	Status
Sprint 1	Repository setup	Development Team	To Do
Sprint 1	User registration	Development Team	To Do
Sprint 1	Login functionality	Development Team	To Do
Sprint 1	Basic UI	Development Team	To Do
Sprint 1	Testing	Development Team	To Do
Sprint 2	Add Student	Development Team	To Do
Sprint 2	Search Student	Development Team	To Do
Sprint 2	Update Student	Development Team	To Do
Sprint 2	Delete Student	Development Team	To Do
Sprint 3	Add Course	Development Team	To Do
Sprint 3	Assign Courses	Development Team	To Do
Sprint 3	Record Attendance	Development Team	To Do
Sprint 3	Attendance Reports	Development Team	To Do

11. Scrum Board

Backlog	To Do	In Progress	Testing	Completed
Login	Add Student	Search Student	Reports	Setup
Course	Attendance	Notifications		
Marks	Update Student			

The Scrum Board should be updated throughout Sprint execution to monitor project progress.

12. Scrum Ceremonies

Ceremony	Activities
Sprint Planning	Select User Stories, estimate effort and define the Sprint Goal.
Daily Scrum	Each member answers: What did I complete yesterday? What will I do today? What challenges am I facing?
Sprint Review	Demonstrate completed software features and collect feedback.
Sprint Retrospective	Discuss what went well, what challenges occurred and what improvements should be implemented.

13. Burndown Chart Plan

The team will track remaining Story Points during each Sprint. A Burndown Chart can be maintained with Sprint Day on the horizontal axis and Remaining Story Points on the vertical axis. The expected trend is a gradual reduction of remaining work toward zero by the end of the Sprint.

Sprint Day	Planned Remaining Points	Actual Remaining Points
Day 1	20	20
Day 2	16	____
Day 3	12	____

Day 4	8	____
Day 5	4	____
Day 6	0	____

14. Sprint Review Report

At the Sprint Review, the team demonstrates completed features to stakeholders and collects feedback. For this project, the review should cover the completed authentication, student management, course and attendance features according to the selected Sprint.

Review Points:

- Features completed during the Sprint
- Features not completed and reasons
- Stakeholder feedback
- Required changes or improvements
- Items to be considered for the next Sprint

15. Sprint Retrospective Report

Area	**Observation / Improvement**
What went well?	Team collaboration, task division and regular progress tracking should be discussed by the team.
Challenges	Identify technical issues, dependencies, unclear requirements or time constraints faced during the Sprint.
What should improve?	Improve estimation, communication, testing and task planning where required.
Action Items	Assign clear owners and deadlines for improvements in the next Sprint.

16. Agile Framework Comparison

Framework	**Main Idea**	**Project Suitability**
Scrum	Uses defined roles, events and artifacts; suitable for iterative project development.	Suitable
Kanban	Uses a visual workflow and focuses on continuous flow of work.	Suitable for continuous maintenance
Extreme Programming (XP)	Emphasizes engineering practices and frequent feedback.	Useful when development quality and frequent changes are priorities
Lean Development	Focuses on reducing waste and improving value delivery.	Useful for process improvement
Large Scale Scrum (LeSS)	Extends Scrum principles for larger development efforts.	Useful for larger teams/products
Scaled Agile Framework (SAFe)	Provides a structured approach for scaling Agile across larger organizations.	Useful for large-scale environments
Scrum@Scale	Applies Scrum principles across	Useful when several Scrum

	multiple Scrum teams.	teams coordinate

Recommendation: Scrum is a suitable framework for this Student Record and Academic Management Platform because the Unit 2 project is organized around Scrum roles, Product Backlog, Sprint Planning, Scrum Board and Scrum ceremonies.

17. Project Timeline and Release Roadmap

Phase	Major Work	Expected Outcome
Phase 1	Product discovery and project setup	Project scope, requirements and repository setup
Sprint 1	Authentication and basic UI	User registration/login and initial interface
Sprint 2	Student Management	Add, search, update and delete student records
Sprint 3	Course and Attendance	Course assignment, attendance recording and reports
Release / Review	Testing, demonstration and feedback	Reviewed increment and improvement plan

18. Conclusion

The Unit 2 Agile Mini-Project converts the Student Record and Academic Management Platform into an Agile development plan. The project is divided into Epics and User Stories, prioritized in a Product Backlog, estimated using Story Points and organized into Sprints. Scrum roles and ceremonies provide a structured way for the team to collaborate, track progress, review completed work and continuously improve the product.

19. Student Submission Checklist

- Agile Project Planning Report
- Product Vision Document
- Scrum Team Structure
- Scrum Roles and Responsibilities Document
- Epic Documentation
- User Story Repository
- Acceptance Criteria Document
- Prioritized Product Backlog
- Story Point Estimation Sheet
- Sprint Planning Document
- Sprint Backlog
- Scrum Board
- Burndown Chart
- Sprint Review Report
- Sprint Retrospective Report
- Agile Framework Comparison Report
- Project Timeline and Release Roadmap
- Complete Agile Documentation